

HERNANDO, PANTINE B.

BSIT-3

PICO CTF- CHALLENGE 7

shark on wire 1

Medium Forensics picoCTF 2019

AUTHOR: DANNY

Description

We found this [packet capture](#). Recover the flag.

23,345 users solved

88%

Liked

picoCTF{FLAG}

Submit Flag

capture.pcap

From <https://play.picoctf.org>

Analysis

- The investigation involved analyzing various protocols within the capture, including SSDP BROWERS, LLMNR, ARP, UDP, and TCP.

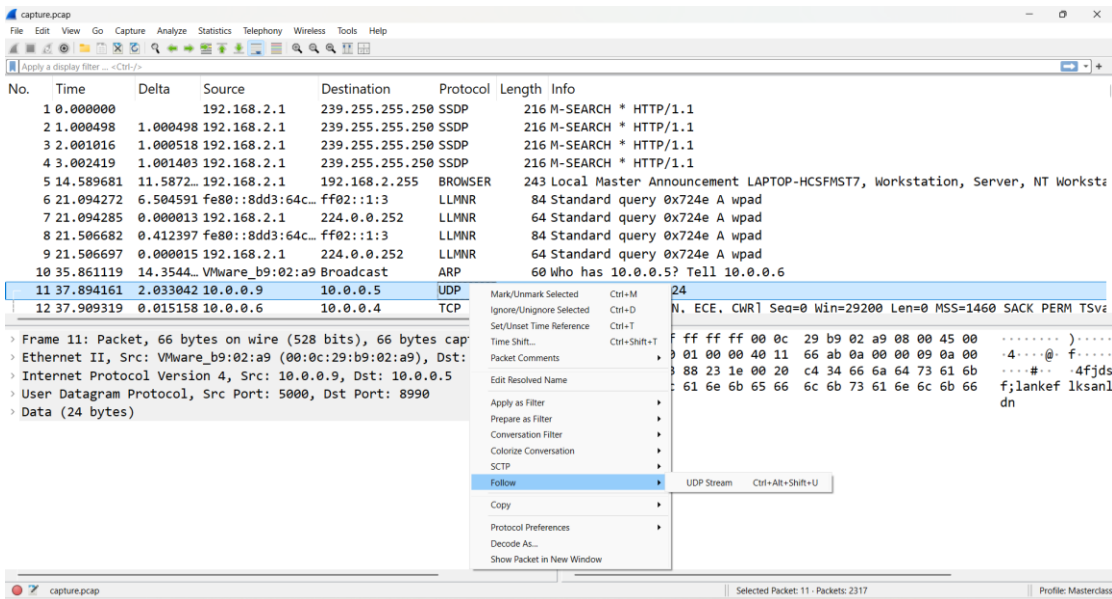
- Protocol Filtering:** While many packets involved standard network noise like "Local Master Announcements" or "Standard queries," the focus shifted to UDP traffic.

- Packet 11 Details:** A significant UDP packet (Frame 11) was identified traveling from source 10.0.0.9 to destination 10.0.0.5.

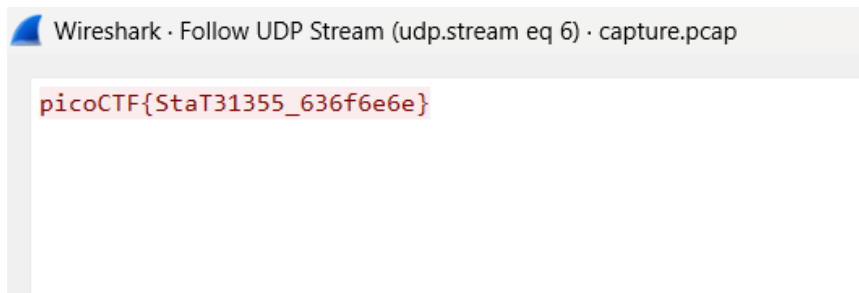
- Port Information:** This specific communication occurred over Source Port **5000** and Destination Port **8998**.

No.	Time	Delta	Source	Destination	Protocol	Length	Info
1	0.000000		192.168.2.1	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
2	1.000498	1.000498	192.168.2.1	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
3	2.001016	1.000518	192.168.2.1	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
4	3.002419	1.001403	192.168.2.1	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
5	14.589681	11.5872...	192.168.2.1	192.168.2.255	BROWSER	243	Local Master Announcement LAPTOP-HCSFMST7, Workstation, Server, NT Worksta
6	21.094272	6.504591	fe80::8dd3:64c...	ff02::1:3	LLMNR	84	Standard query 0x724e A wpad
7	21.094285	0.000013	192.168.2.1	224.0.0.252	LLMNR	64	Standard query 0x724e A wpad
8	21.506682	0.412397	fe80::8dd3:64c...	ff02::1:3	LLMNR	84	Standard query 0x724e A wpad
9	21.506697	0.000015	192.168.2.1	224.0.0.252	LLMNR	64	Standard query 0x724e A wpad
10	35.861119	14.3544...	VMware_b9:02:a9	Broadcast	ARP	60	who has 10.0.0.5? Tell 10.0.0.6
11	37.894161	2.033042	10.0.0.9	10.0.0.5	UDP	66	5000 -> 8998 Len=24
12	37.909319	0.015158	10.0.0.9	10.0.0.4	TCP	74	48520 -> 8000 [SYN, ECE, CWR] Seq=0 Win=29200 Len=0 MSS=1460 SACK PERM TSv

Frame 1: Packet, 216 bytes on wire (1728 bits), 216 bytes captured (1728 bits) on interface 0
Ethernet II, Src: 0a:00:27:00:00:08 (0a:00:27:00:00:08), Dst: IPv4mcast
Internet Protocol Version 4, Src: 192.168.2.1, Dst: 239.255.255.250
User Datagram Protocol, Src Port: 52688, Dst Port: 1900
Simple Service Discovery Protocol



• **Stream**
Extraction: The flag was successfully isolated by using the Wireshark "Follow UDP Stream" feature specifically for UDP stream 6.



• **Flag** **Recovered:**
 picoCTF{StaT31355_636f6e6e}

Reflection:

This forensics challenge, highlights the importance of protocol analysis in cybersecurity. While the capture.pcap file contained various types of network noise like SSDP and ARP traffic, the key to the solution was isolating specific communication channels. By filtering for **UDP Stream 6** in Wireshark, the hidden data was successfully extracted from the background traffic. This process demonstrates how flags can be concealed within common transport layer protocols, requiring a methodical approach to stream reassembly to recover the final string: picoCTF{StaT31355_636f6e6e}.

